

# Graph based clustering for two-tier architecture in Internet of things

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**Abstract**—The realization of IoT in real world without sensing devices is impossible. These sensing devices are battery operated and energy constrained in terms of computation and communication. The optimized communication leads to longer network life time. Minimum hop count, increased connectivity and scalability are challenging issues that can be effectively addressed by clustering mechanism. In the two tier architecture of IoT, namely IoT layer and sensing layer, to have minimum hop communication and connectivity among the devices, we utilize two well known graph theory concepts namely dominating set and bipartite graph. The device communication is modeled using bipartite graph theory for network coverage and connectivity. Dominating set is used for clustering the devices. Test cases have been considered using the SocNetV simulator and evaluated on IoT platform.

**Keywords:** Dominating set, Bipartite Graph theory, Clustering, Internet of Things

## I. INTRODUCTION

Internet of Things (IoT) has grown into multiple dimensions, which encompasses several networks of computers, applications, devices, as well as virtual objects and physical, referred to as objects or things. These objects or things are connected together by utilizing communication technologies such as RFID, mobile networks, wired and wireless, GPS systems, Bluetooth and other evolving technologies. IoT enables innovative services for common citizen, business in industries and government bodies that goes beyond conventional IT services and telecommunication. IoT can acts as a global network of interconnected devices or objects, that enables the discovery and identification of devices along with meaningful processing of data via content, connectivity, computing, context, cognition, collaboration, anytime, anywhere, any service, any network, any object and any human [1], [2].

It is predicted that the approximate devices used currently is 9 billion, and the rate suggest the number of interconnected devices to cross 24 billion by 2020 [1], [3]. Of these devices, an important integral and significant part are sensing and monitoring devices. However, they are energy constrained due to which the power consumption becomes critical aspect for network lifetime. Besides, if these low power devices are equipped with IP, the energy in these devices can be discharged rapidly [4]. Assuming, such huge number of devices are going to communicate and collaborate with other devices in-order to accomplish a given job, effective connectivity among these devices and efficient communication in an optimized way is a non trivial and challenging task.

In this paper, we propose dominating set graph theory based clustering mechanism for two tier architecture of IoT [2], [5]. Hence, we address connectivity issues between the IoT layer and sensing layer by using bipartite graph theory. Dominating set graph theory based clustering mechanism is applied among the devices in sensing layer to have minimum hop communications. Moreover, clustering helps to address additional challenging issues like reduced delay, increased connectivity, minimal energy, maximal network longevity and scalability. Further, we carried out the experiments in IoT and social network based simulators called Cooja and SocNetV.

Rest of the paper is presented as follows: Proposed approach with system model, graph modeling and algorithm using graph theory are described in Section 2. Section 3 explains simulation results. Section 4 concludes the paper with references at the end.

## II. PROPOSED APPROACH

In this section, system model, graph based modeling and terminologies related to our proposed approach of two tier architecture for IoT are as explained below.

### A. System Model

Our system consists of two tier architecture which has been proposed earlier in [2], [5], namely IoT layer and sensing layer. The advantage of these two layer architecture is, practically possible to bring ample applications of IoT [6] in the real world. Because, the devices in IoT layer are mobile IP enabled with IoT standard protocol stack[7], such as IEEE 802.15.4-2006 [8] in physical layer, IEEE 802.15.4e [7] in MAC layer, 6LoWPAN [9] in network layer and in application layer with CoAP protocol [10]. These devices in IoT layer are named as IoT nodes, which are expected to have long battery life, capable of doing real time processing and communication. As shown in Fig 1, the devices in IoT layer are IP enabled referred as  $I = \{a, b, c, d\}$ , which is indicated in blue lines and these nodes can communicate with all other IoT nodes. We consider that the nodes in IoT layer acts as mobile base station because in the traditional approaches [11]–[25], energy discharges rapidly in the nodes that are one-hop away from a base station, while compared to other nodes in the network. Instead of multiple-hop communications, with single or at maximum two hops any node in the sensing layer can reach IoT node irrespective of sensing node position in the network.

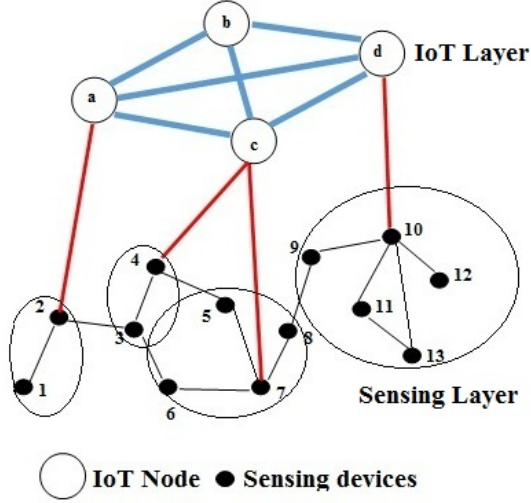


Fig. 1. Two Tier Architecture of IoT

Sensing layer is deployed with devices either IP enabled or ID enabled depending on the requirement of the application. However, we assume that the devices in this layer are ID-enabled such as sensors, actuators, RFID devices, video surveillance devices, cellular mobile phones, and etc. Cluster head (CH) selection and dynamic cluster formation are carried out in this layer. As shown in Fig 1, sensing devices that are referred as  $SD = \{1, 2, 3, \dots, 13\}$  are used to sense and gather the information and send it to the CHs. Devices that are elected as CHs using dominating set mechanism, communicates the received information to IoT nodes. Hence the advantage of our approach is, all the CHs will communicate with one-hop to IoT nodes and all the cluster members communicate one-hop to CHs. We are assuming the nodes are mobile [21], [26], that implies frequent changing of the cluster formation in regular interval periods.

The communication between the devices in the IoT layer to that of sensing layer consists of different possible ways. For instance, one CH might communicate with one base station, two CHs might communicate with one base station and one CH might communicate with two base stations depends whether the base station is within its range of transmission, that is indicated with red lines as shown in Fig 1. This effects every node in sensing layer is capable of understanding the scenario of entire network by communicating with IoT nodes. For instance, the nodes that are elected as CHs referred as  $C = \{2, 4, 7, 10\}$  forms a dominating set of the graph because these vertices covers all the remaining nodes. This strategy allows us to have overall minimum number of communication required among the devices. Further, we considered the residual energy of the nodes in the selection of dominating set. Hence, if these nodes are selected as CHs in the network, the communication for information gathering from sensors and sending to control them can be uniformly distributed to each and every vertex in just one-hop.

## B. Graph based modeling

The graph based terminologies is presented which allows to understand how it is exploited for communication across two layers and formation of clusters.

The bipartite graph theory [27] is used to represent the communication between IoT nodes in IoT layer and cluster heads in sensing layer. A bigraph or bipartite graph is a graph, such that the vertices are divided into two disjoint set of classes, say  $I$  and  $C$  and that are each independent sets, such that every edge connects a vertex in  $I$  to one in  $C$ , where  $I$  is set of IoTnodes i.e.,  $I = \{a, b, c, \dots\}$  and  $C$  is set of cluster heads i.e.,  $C = \{1, 2, \dots, n\}$ . The bipartite representation of communication between the layers enables to address few issues like optimized resource management across the network, collaborative processing and event detection using matching pattern in sensing layer to IoT layer.

Assume  $G = (V, E)$ , an undirected graph representing a network where the edges are the communication links and the vertices are the nodes to define clustering in the network. The clustering mechanism initially divides  $V$  into a collection of subsets  $\{V_1, V_2, \dots, V_k\}$  but not inevitably disjoint, where  $V = \bigcup_{i=1}^k V_i$ , such that each subset  $V_i$  induces a connected subgraph of  $G$ . Overlapping of these these induced subgraphs could be possible. Then, every such subset vertex is a cluster. Let us say,  $G = (V, E)$  is an abstracted graph is constructed, such that for every vertex  $v_i \in V$  represents to a subset  $V_i$ . In general, in each cluster a particular vertex is selected that acts as cluster head or cluster leader and that corresponding subset forms the cluster.

In a graph  $G = (V, E)$ , a set  $S \subseteq V$  of vertices is a dominating set ( $\alpha$ -set), if every vertex  $v \in V$  is adjacent to an element of  $S$  or an element of  $S$ . A  $\alpha$ -set  $S$  of a graph  $G$  is a minimal dominating set, if and only if for any  $x \in S$ , implies there is  $v \in (V - S)$  for which  $N[v] \cap S = x$  or  $x$  is an isolate of  $S$ . Let  $S$  be a  $\alpha$ -set of  $G$ . Then, for every vertex  $x \in S$ ,  $S - x$  is not a  $\alpha$ -set of  $G$ . Thus, there is a vertex  $v \in (V - S) \cup x$ , that is not dominated by any vertex in  $S - x$ . Now, either  $v = x$ , which implies  $x$  is an isolate of  $S$  or  $v \in V - S$ , in which case  $v$  is not dominated by  $S - x$ , and is dominated by  $S$ . This shows that  $N[v] \cap S = x$ . In order to prove the converse, we assume  $S$  is a  $\alpha$ -set and for all  $x \in S$ , either there is  $v \in (V - S)$  for which  $N[v] \cap S = x$  or  $x$  is an isolate of  $S$ . We assume to the contrary that  $S$  is not a  $\alpha$ -set( $G$ ). Thus, there is a vertex  $x \in S$  such that  $S - x$  is a  $\alpha$ -set of  $G$ . Hence,  $x$  is adjacent to at least one vertex in  $S - x$ , so condition  $x$  is an isolate of  $S$  does not hold. Also, if  $S - x$  is a  $\alpha$ -set, then every vertex in  $V - S$  is adjacent to at least one vertex in  $S - x$ , implies the condition  $v \in (V - S)$  for which  $N[v] \cap S = x$  does not hold for  $x$ .

## C. Proposed clustering algorithm

For the proposed method as stated dominating set and bipartite graph are used and the details of algorithm is described as follows. Dominating set is NP-complete decision problem [28], but many approximation algorithms exists which gives solutions optimal up to a certain factor. A simple approach

is using greedy strategy by selecting a node  $u$ , that has highest degree i.e.,  $\Delta(G)$  with maximum residual energy ( $RE$ ). Then, by finding  $N(u)$ , to remove the neighbors from the set which bifurcates the covered nodes from the uncovered nodes. Again, the same strategy is carried till all nodes are covered. the selected nodes of  $u$  are considered as dominating set that acts as CHs. The natural greedy approach works is given in the following algorithm. As a part of nodes uniform energy consumption, we are avoiding the same node to be elected as CH again and again using mobility. Besides, this is the significant reason for using residual energy as another parameter to be elected as CH that by default comes into the dominating set. The algorithm overall time complexity coverages to  $O(V)$  where,  $V$  is the vertices or nodes.

**Algorithm 1:** Greedy algorithm for dominating set

**Data:** Graph  $G(V, E)$   
**Result:** DS (Dominating set)

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1 for each  $v \in V$  do
2    $u = \text{Select}(\Delta(G), \text{Max}(RE))$  // where  $\Delta(G)$  is
   maximum degree and  $RE$  is residual energy;
3   Find  $N(u)$  // Where  $N(u)$  indicates is neighbors
   of node  $u \{u|u \in G\}$ ;
4    $DS = DS \cup u$  // where  $DS$  indicates dominating
   set and adding  $u$  into set ;
5    $V = \text{update}(V) \{V|V \notin N(u), u\}$  // Remove
   from the covered nodes from the existing vertices
6 end
7  $CHs = DS$  // Dominating set nodes acts as CHs.
8
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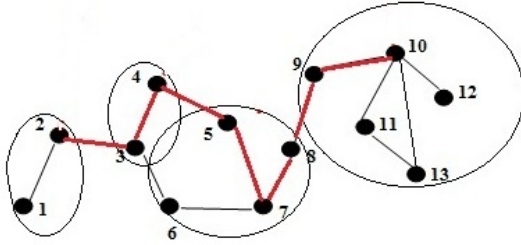


Fig. 2. Connected Dominating Set

If you connect all the vertices in the dominating set then it is called connected dominating set as shown in Fig 2. For routing computation in the network, connected dominating sets are helpful. This routing application can be achieved by using a small connected dominating set that acts as a backbone for communications. For example, nodes 2, 3, 4, 5, 7, 8, 9, 10 acts the routing path of the network. The remaining nodes that are not in the set of connected path, can have the communication by passing messages through neighbors.

### III. SIMULATION RESULTS

The proposed method is evaluated using IoT based simulator. For better understanding, we compare our hierarchical two tier architecture with flat architecture based clustering

algorithms. For this purpose, we utilize the most widely used clustering approach LEACH [11]. The simulation results of the proposed approach is carried out in IoT based Cooja simulator [29]. We deployed the network with three types devices viz., sky mote, wismote and Z1mote which are virtual motes available in the simulator. The sensing information like temperature and light intensity values from sensing layer are communicated to IoT layer as shown in the Fig 3. As we stated in the above section, IoT layer consists of 6 IP devices and in sensing layer sensor devices are deployed. By giving the IPv6 address of IoT in the web browser, we can able to analyze the average sensing information across the scattered network from corresponding IoT nodes.

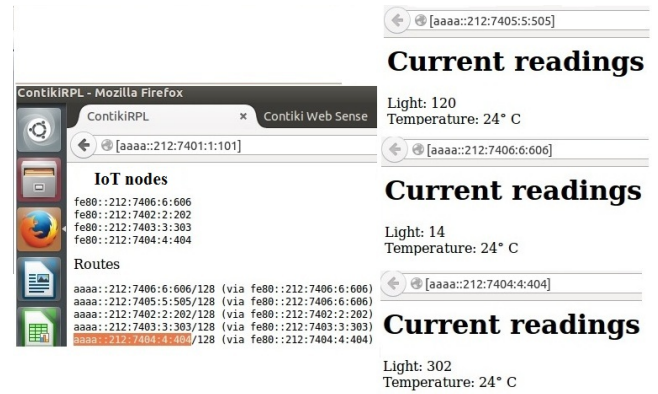


Fig. 3. Sensing Values of the network in Layer1

The comparative results of power consumption with LEACH algorithm and our proposed algorithm is given as Fig 4. The results shows that the power consumption of each node and our proposed algorithm consumes less power as compared with LEACH.

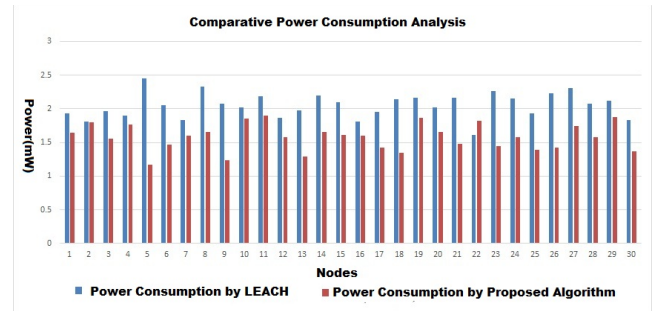


Fig. 4. Comparative Analysis of Power Consumption

Since, dominating set promises the minimum selection of CHs, helps to prolong the network lifetime. Also, the proposed algorithm has taken into consideration the CH selection based on residual energy, the stability in the network is preserved and helps to extend the whole network lifetime. The comparative analysis of CHs selection of leach and proposed algorithm is shown in the Fig 5.

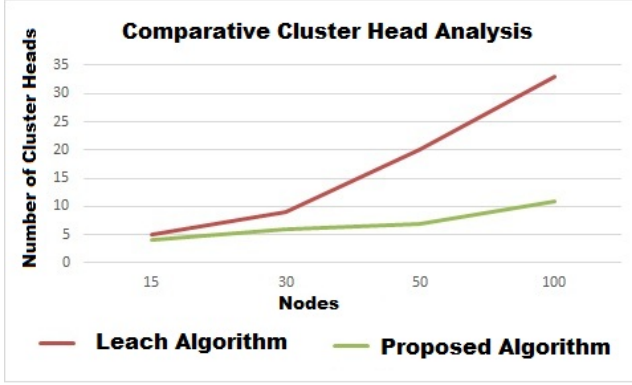


Fig. 5. Number of Cluster Head's Comparison

We evaluated our algorithm in terms of communication cost by comparing flat leach topology and our proposed layered topology. Since, mobile base station can be able to move to any place, CH communicates with it in a single hop. However, in fixed BS, CH has to communicate in a multi hop fashion. The comparative analysis of hops between leach and proposed algorithm is shown in Fig 6.

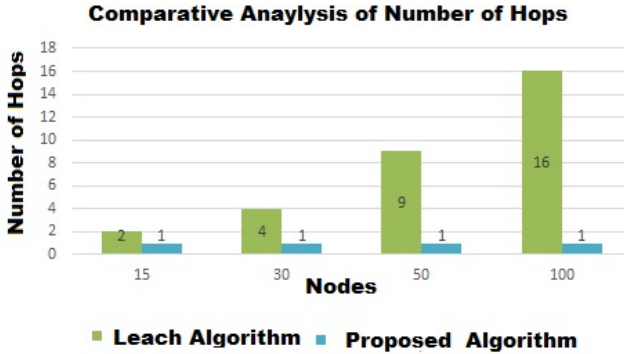


Fig. 6. Comparison of Number of Hops

The communication cost of layered topology is low when compared with flat topology as shown in Fig 7. Also, in WSN more than 50% of cost is due to communication [30], our proposed approach is giving better performance. Further, the stability in the network depends on the battery of the sensors, more the communication sooner the network will become unstable. Hence, our approach is more stable.

#### IV. CONCLUSION

Dominating set theory is used for clustering the nodes and modeled using bipartite graph theory for two tier architecture of IoT. It promises and reasons out as suitable architecture for various applications that are assured by IoT. The algorithm elects the minimum CHs given a network and forms a cluster to have efficient communication. The crucial factor of the network depends on the residual energy of the nodes were

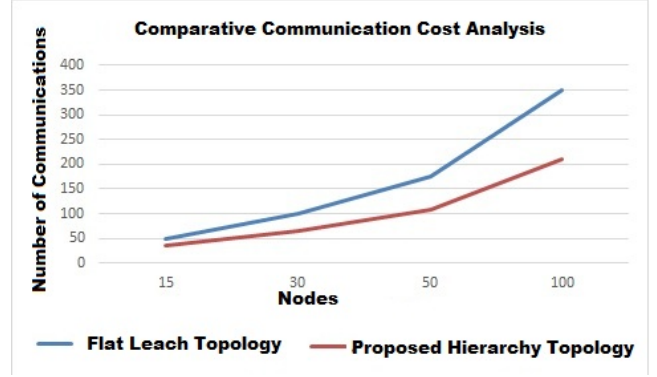


Fig. 7. Comparative Analysis of Communication cost

taken into consideration for cluster formation, reasons out as a simple approach but dynamic and efficient. Further, the algorithm implementation has been done using IoT based cooja simulator and Socvnet. Results show that the optimal number of CH has been elected and thus indicating maximum network coverage, better communication cost and the amount of power consumption in the network promises prolong network lifetime. Further, comparative analysis with the basic LEACH clustering algorithm and our proposed reasons out as suitable better approach for IoT.

#### V. ACKNOWLEDGMENTS

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